

Volume 43

Sentinel Identity Security Engine (SISE)

Status: Core Security Domain

Priority: Critical

Vision

Project Sentinel does not merely manage accounts.

It understands the complete identity ecosystem.

Every person.

Every service account.

Every API identity.

Every machine identity.

Every certificate.

Every token.

Every relationship.

Identity becomes a Living Object inside the Digital Twin.

Identity Philosophy

Identity is no longer just:

Username

Password

Instead identity consists of:

Person

Everything is connected.

Identity Object

Each identity becomes a Sentinel Object.

Example

PS-ID-000812394

Supported Identity Types

The engine supports:

- Human identities
- Service accounts
- Machine identities
- API identities
- Cloud identities
- Privileged identities
- Guest users
- External partners
- Temporary identities
- Break-glass accounts
- Certificate-based identities
- Device identities

Each follows the same object model.

Identity Lifecycle

Created

Every transition becomes an event.

Identity Relationships

Example

Employee

This relationship graph is extremely valuable during investigations.

Authentication Evidence

Examples

Password Status

MFA Status

Passkeys

FIDO2 Keys

Certificates

Authenticator Apps

Recovery Codes

Recovery Email

Recovery Phone

Identity Provider

Federation

Session Policies

Everything becomes structured evidence.

Authorization Evidence

Examples

Role Membership

Administrator Rights

Group Membership

Cloud Roles

Application Roles

API Permissions

Database Roles

Temporary Elevation

Every permission is evidence.

Identity Health

Identity health is composed of:

Authentication

Authorization

Recovery

Verification

Policy Compliance

Session Security

Device Trust

Evidence Freshness

Trust

Trust Calculation

Identity trust is based on:

Verified MFA

Verified Devices

Verified Recovery

Verified Authentication

Permission Reviews

Policy Compliance

Relationship Confidence

Evidence Freshness

No hidden logic.

Identity Timeline

Example

Identity Created

Everything is preserved.

Digital Twin Integration

Every identity connects to:

Devices

↓

Applications

↓

Cloud

↓

Certificates

↓

Email

↓

VPN

↓

Business Processes

↓

Departments

↓

Projects

Identity becomes one of the central nodes in the Knowledge Graph.

Home User Dashboard

The platform answers:

- Are all my important accounts protected?
 - Is MFA enabled where available?
 - Are my recovery options complete?
 - Which devices are linked?
 - Which sessions are active (where supported)?
 - What should I improve next?
-

Enterprise Dashboard

Administrators can view:

Identity Coverage



MFA Adoption



Privileged Accounts



Service Accounts



Dormant Accounts



Risk Trends



Verification Status

NEW CORE ENGINE

Identity Relationship Engine

Maps relationships between:

- Users
- Devices
- Applications
- Certificates
- Cloud accounts
- API keys
- Service accounts
- Business roles

This enables impact analysis and governance.

NEW CORE ENGINE

Identity Verification Engine

Verifies:

- Ownership
- Enrollment
- MFA configuration
- Device association
- Recovery readiness
- Identity lifecycle state

Verification results become evidence.

NEW CORE ENGINE

Identity Hygiene Engine

Continuously checks for:

- Dormant accounts
- Unused service accounts
- Stale permissions
- Excessive privilege
- Duplicate identities
- Missing recovery methods

The engine produces recommendations rather than assumptions.

NEW CORE ENGINE

Session Intelligence Engine

Where supported by integrated identity providers and enterprise systems, the platform can observe:

- Active sessions
- Session duration
- Geographic patterns
- Device associations
- Authentication methods

This supports investigation and governance while respecting privacy and platform capabilities.

NEW IDEA

Identity Passport

Every identity receives a continuously updated passport.

Example

Identity

NEW IDEA

Identity Journey

Instead of isolated settings.

Users see their identity evolve.

Example

January

Security becomes a journey rather than a checklist.

NEW IDEA

Identity Guardian

One-click identity review.

For home users:

"My Accounts"

↓

Google

Microsoft

Apple

GitHub

Bank

Crypto Exchange

Email

Social Media

The Guardian helps users verify protection status and improve account security through guided recommendations. Integrations depend on the capabilities and APIs provided by each service.

NEW IDEA

Identity Dependency Map

One of our strongest enterprise capabilities.

Example

Employee

Executives can immediately understand operational impact if an identity changes or is unavailable.

PS-ADR-0038

Identity-Centric Security

Status: Accepted

Decision

Identity shall be treated as a primary security domain with dedicated object modeling, relationship management, trust evaluation, lifecycle tracking, and evidence collection.

Rationale

Modern cyber attacks frequently target identities and access paths. A comprehensive identity model strengthens governance, visibility, and response across the entire platform.

NEW LONG-TERM VISION

Unified Identity Fabric

As our architecture matures, I think we should define another foundational concept:

The **Unified Identity Fabric**.

Instead of separate identity systems for endpoints, cloud, mobile, APIs, certificates, and applications, Project Sentinel creates one coherent identity layer that spans the entire Digital Twin.

Every authenticated interaction becomes understandable, traceable, and explainable through relationships and evidence.

A new flagship capability

Sentinel Identity Confidence

Rather than asking only:

"Is this user using MFA?"

Project Sentinel asks:

- Is this identity well protected?
- Is it consistently verified?
- Are its recovery methods healthy?
- Are its trusted devices current?
- Are its permissions appropriate?
- Is its security improving over time?

The result is an **Identity Confidence Profile**, built entirely from verifiable evidence—not assumptions.

This profile will help home users, enterprises, financial institutions, healthcare organizations, governments, and other sectors understand identity security in a clear and measurable way.

Looking Ahead

The next volume will be **Volume 44 - Sentinel Cloud Security Engine (SCSE)**.

This will extend the same object-centric architecture into cloud environments, modeling accounts, subscriptions, projects, virtual machines, storage, networking, serverless functions, identities, containers, and managed services as Living Objects within the Digital Twin. Together with the Identity Security Engine, it will provide a unified view of modern hybrid environments while remaining faithful to our principles of evidence-first analysis, explainability, accessibility (WCAG), multilingual support, and modular Python implementation.

=====[USER]=====

Good and continue

=====[ASSISTANT]=====

Excellent. We have now arrived at one of the largest enterprise modules in Project Sentinel.

Cloud has fundamentally changed cybersecurity.

There is no longer a clear network perimeter.

Instead, security revolves around:

- Identities
- Resources
- Relationships
- Policies
- Workloads
- Data

Fortunately, our architecture is already designed for this.

Everything becomes a **Sentinel Object**.

That means we don't need a separate cloud architecture.

We extend the same Object Kernel.

PROJECT SENTINEL BLUEPRINT

Volume 44